

ERP / PMO Intelligence Dashboard

From Project Reporting to Management Decision Support

Project Overview

ERP implementation projects generate a large amount of operational and managerial data: project progress, schedule variance, risks, issues, backlog, SLA performance, actions, decisions, meetings, and customer-related information.

The challenge is not simply to report these numbers.

The real question is:

How can fragmented PMO data be transformed into an executive-level decision-support system?

This project presents a Power BI Executive PMO Intelligence Dashboard designed to connect different dimensions of ERP project performance and provide management with a consolidated view of portfolio health, risk, operational pressure, and governance.

The analytical journey follows:

Reporting → Monitoring → Diagnosis → Early Warning

Business Objective

The primary objective is to provide a unified view of ERP project portfolio performance and help management answer four key questions:

1. **What is happening across the project portfolio?**
2. **Which projects require attention?**
3. **What operational or governance signals are driving project risk?**
4. **Which projects may be deteriorating and require early intervention?**

Instead of looking at project indicators independently, the dashboard connects:

Risk + Stability + Progress + Schedule + Issues + Backlog + SLA + Actions + Decisions

Dashboard Structure

The dashboard is organized around an executive analytical story.

01 — Executive Overview

Provides a high-level view of the current project portfolio.

Key indicators include:

- Total Projects
- Stable Projects
- Stabilizing Projects
- Unstable Projects
- Critical Issues
- Backlog
- Overdue Actions
- Risk Score
- SLA Compliance
- Risk Trend

Business Question:

What is the current health of the ERP project portfolio?

02 — Risk & Project Performance

Focuses on identifying where project risk is concentrated.

Key dimensions include:

- Risk Score
- Stability
- Project Progress
- Schedule Variance
- SLA Compliance
- Project Phase
- Strategic Indicators

Business Question:

Which projects are currently carrying the highest risk?

03 — Operational Performance

Connects project performance with operational signals.

The analysis includes:

- Ticket Volume
- Issues
- Critical Issues
- Backlog
- SLA
- Root Cause Categories
- Issue Ownership

Business Question:

Are operational indicators showing signs of project deterioration?

04 — Governance & Management Actions

Project performance is also influenced by management decisions and follow-up actions.

This page provides visibility into:

- Pending Decisions
- Overdue Actions
- Action Priority
- Open Issues
- Project Priority
- Meeting Activity

Business Question:

Where is management intervention required?

05 — Project Health

A composite health framework combines multiple dimensions of project performance.

The health model considers:

- Progress Health
- Schedule Health
- Risk Health
- Issue Health
- Backlog Health
- SLA Health
- Change Health
- Action Health
- Decision Health

These indicators are combined into a **Project Health Score** to provide an executive-level signal for prioritizing management attention.

Business Question:

How healthy is the project when multiple performance dimensions are considered together?

Early Warning — Next Development Stage

The next stage of the project is to extend the dashboard from monitoring toward **Early Warning**.

The key shift is from:

Which projects are risky?

to:

Which projects are getting worse?

The Early Warning layer will analyze changes over time, including:

- Risk Change WoW
- SLA deterioration
- Backlog growth
- Stability changes
- Operational pressure

Projects can then be classified into warning levels such as:

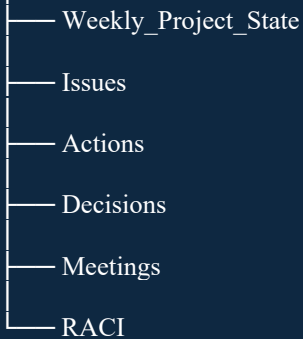
Critical → High → Medium → Normal

This creates a rule-based early warning mechanism before moving toward Machine Learning.

Data Model

The analytical dataset is structured around several PMO entities:

Project_Master



Dim_Date



The central analytical table is:

Weekly_Project_State

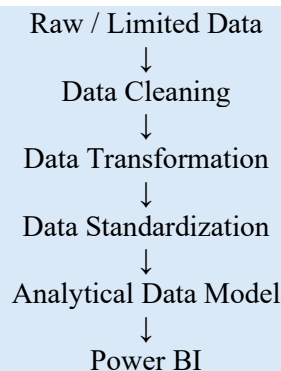
This table captures the project condition over time and enables:

- Weekly monitoring
- Latest-state analysis
- Week-over-Week comparisons
- Risk trend analysis
- Project health calculation
- Future predictive modeling
- Potential temporal TDA analysis

Data Preparation

Data preparation was an important part of the project.

The workflow included:



Tools Used

- **Power Query** for data transformation and preparation
- **DAX** for analytical calculations and KPI logic
- **Power BI** for data modeling and dashboard development

Dataset Development & Confidentiality

The initial project started with a **limited amount of real-world data**.

To create a sufficiently rich dataset for analytical development, the original sample was expanded using **AI-assisted prompting** to generate additional synthetic records while maintaining the required data structure and analytical characteristics.

To protect confidentiality, the next stage involved replacing key information and metadata with **fictional and anonymized values**, including project, customer, and organizational identifiers.

Therefore, the final dataset is designed to preserve the **analytical structure and business logic** of the original context without exposing confidential information.

The dataset should be considered synthetic/anonymized and is intended for analytical, educational, portfolio, and demonstration purposes.

KPI Framework

The dashboard contains several groups of KPIs.

Portfolio

- Total Projects
- Stable Projects
- Stabilizing Projects
- Unstable Projects
- Unstable Project %

Risk

- Average Risk Score
- Risk Change WoW
- Risk Health

Schedule

- Delayed Projects
- Average Schedule Variance
- Progress Health
- Schedule Health

Operations

- Open Issues
- Critical Issues
- Backlog
- SLA Compliance

Governance

- Pending Decisions
- Overdue Actions
- Decision Health
- Action Health

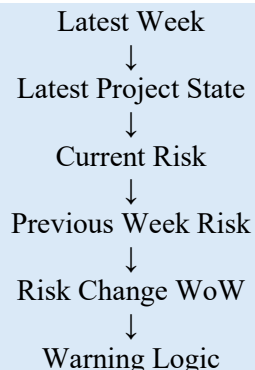
Composite

- Project Health Score
- Project Health Status

DAX & Analytical Logic

DAX was used to transform raw project-state data into business-oriented KPIs.

Examples of analytical logic include:



This temporal structure is particularly important because a static Risk Score does not always tell the complete story. For example:

A project with moderate risk but rapidly increasing risk may deserve more attention than a project with higher but stable risk.

💡 Data Storytelling

The dashboard was designed around a management narrative rather than simply displaying a collection of charts. The analytical story moves from:



Early Warning

This approach aims to transform the dashboard from a **reporting tool** into a **decision-support layer**.

🗺️ Future Analytical Roadmap

The project is designed to evolve through several analytical stages.

Phase 1 — Descriptive Analytics

Portfolio reporting and KPI monitoring.

Phase 2 — Diagnostic Analytics

Project health, trend analysis, and root-cause exploration.

Phase 3 — Early Warning

Rule-based detection of deteriorating projects.

Phase 4 — Predictive Analytics

Machine Learning models for predicting future risk escalation.

Phase 5 — Topological Risk Analysis

A research-oriented extension based on **Topological Data Analysis (TDA)**.

The proposed research question is:

Can topological changes in the multidimensional project-state space reveal increases in ERP project risk earlier than conventional PMO indicators?

A weekly project state can potentially be represented as a multidimensional vector:

Project State =

[
Progress,
Schedule,
Risk,
Issues,
Backlog,
SLA,
Change,
Resource Risk,
Customer Risk,
Data Risk
]

The collection of project states at each point in time can then be interpreted as a temporal point cloud.

This creates a potential bridge between:

PMO Analytics → Machine Learning → Topological Data Analysis

🔧 Skills Demonstrated

Business & Analytics

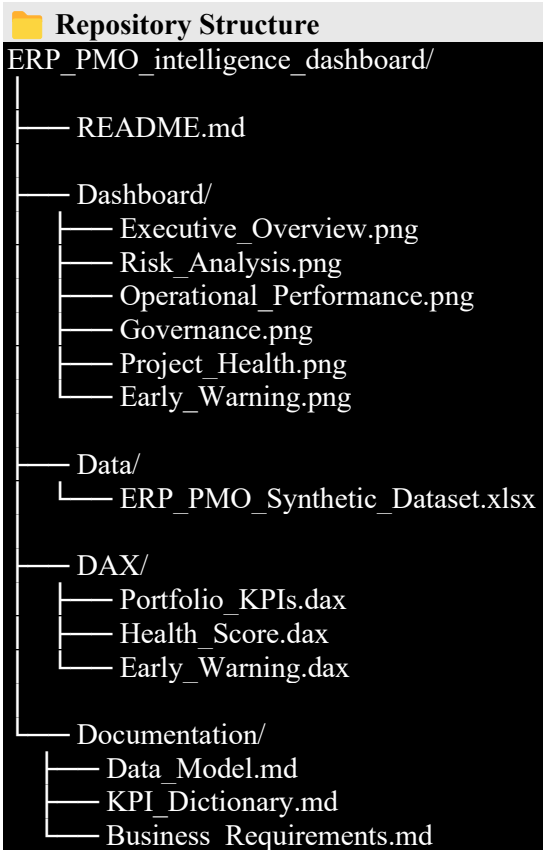
- Business Analysis
- PMO Analytics
- ERP Project Analytics
- KPI Design
- Risk Analytics
- Data Storytelling
- Decision Support

Technical

- Power BI
- Power Query
- DAX
- Data Cleaning
- Data Transformation
- Data Modeling
- Dashboarding
- Time-based Analysis

Future / Research

- Machine Learning
- Early Warning Systems
- Topological Data Analysis
- Temporal Data Analysis



Key Takeaway

The central idea behind this project is simple:

A good PMO dashboard should not only explain what happened. It should help management understand where attention is needed and, eventually, what may happen next.

The project therefore represents a progression from:

Data → KPI → Insight → Decision → Early Warning

rather than simply:

Data → Visualization

Project Status

Component	Status
Data Cleaning	● Completed
Data Transformation	● Completed
Data Model	● Developed
KPI Framework	● Developed
Executive Dashboard	● Completed
Project Health	● Developed
Early Warning	● Next Stage
Machine Learning	○ Planned
TDA / Topological Risk	○ Research Stage

Disclaimer

This repository uses synthetic/anonymized data developed for analytical and portfolio purposes. No confidential organizational, customer, project, or personal information is intentionally exposed.